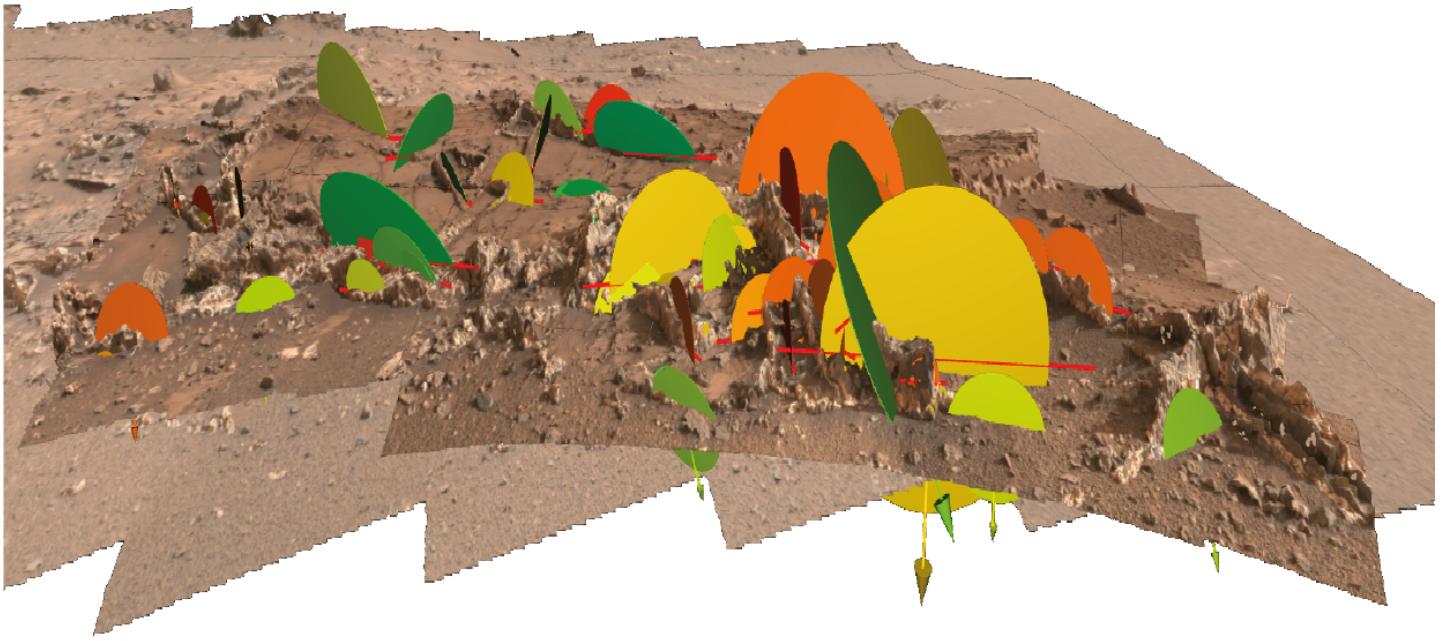
PRo3D 2.9.4

User Manual



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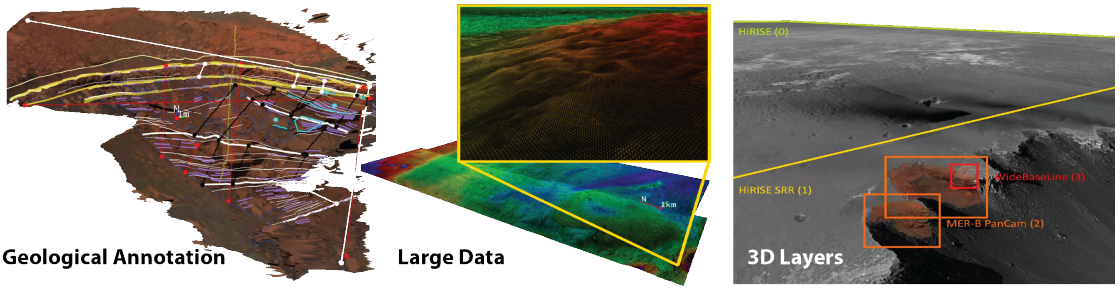
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1 Introduction



PRo3D, short for **P**lanetary **Ro**botics **3D** Viewer, is an interactive 3D visualization tool to allow planetary scientists to work with high-resolution 3D reconstructions of the Martian surface.

1.1 Who uses PRo3D

PRo3D aims to support planetary scientists in the course of NASA's and ESA's missions to find signs of life on the red planet by exploring high-resolution 3D surface reconstructions from orbiter and rover cameras.
For the past 5 years the development of PRo3D has been geared towards providing planetary **geologists** with interactive tools to digitize geological features on digital outcrop models (DOMs) on the Martian surface. During our fruitful cooperation with geologists from the Imperial College of London, PRo3D has emerged as their main tool to conduct remote geological analysis which lead to many publications and talks at various geological science venues.
Planetary geology is the most elaborately supported use-case of PRo3D, however we strive to expand our user groups to other use-cases, so we have also developed features for supporting science goals in **landing site selection** and **mission planningabc**.

1.2 Features

- **Geological Annotation**: PRo3D lets users pick points on the 3D surface at the full resolution of the data present. Our tools contain point, line, and polyline annotations, while line segments are projected onto the surface. Various measurements are computed at the highest possible accuracy, such as the distance along a 3D surface (waylength) or dip-and-strike orientations of sediment structures.

- **Large Data**: Surface reconstructions from high-resolution satellite images can easily yield gigabytes of data in terms of geometry, imagery, and additional layers. With PRo3D users can explore huge datasets interactively and even perform measurements of topographic features. The displayed

dataset on the right consists of 2GB of raw 3D position vectors, a 1GB elevation map, and 10GB of image data rendered at interactive framerates with commodity hardware, utilizing adjustable level-of-detail and out-of-core techniques.

- **3D Layers**: Although, PRo3D is not a GIS system, we need to provide our users with typical GIS features to solve their geospatial problems, such as evaluating topographic or geological features. Our 3D layering technique allows a seamless integration of different reconstructions present at a single location. Unlike image or DTM layering we allow users to blend full 3D data by assigning rendering priorities, which is crucial to explore reconstructions from multiple rover camera instruments.

1.3 Data

Currently, PRo3D only supports reconstructions in the proprietary data format OPC (Ordered Point Clouds), basically consisting of hierarchically organized surface patches. These reconstructions stem from orbiter images and rover images and are produced by Joanneum Research by using the PRoViP processing pipeline. Many surface reconstructions have been generated from, for instance: HiRISE, MER-A, MER-B and MSL missions from various instruments. An ongoing project evaluates terrestrial applicability of PRo3D and the PRoViP pipeline by capturing outcrops in the UK.

2 Start

Start the viewer by clicking the PRo3D.exe and open the Menu in the top left of the window, shown in Figure 1.

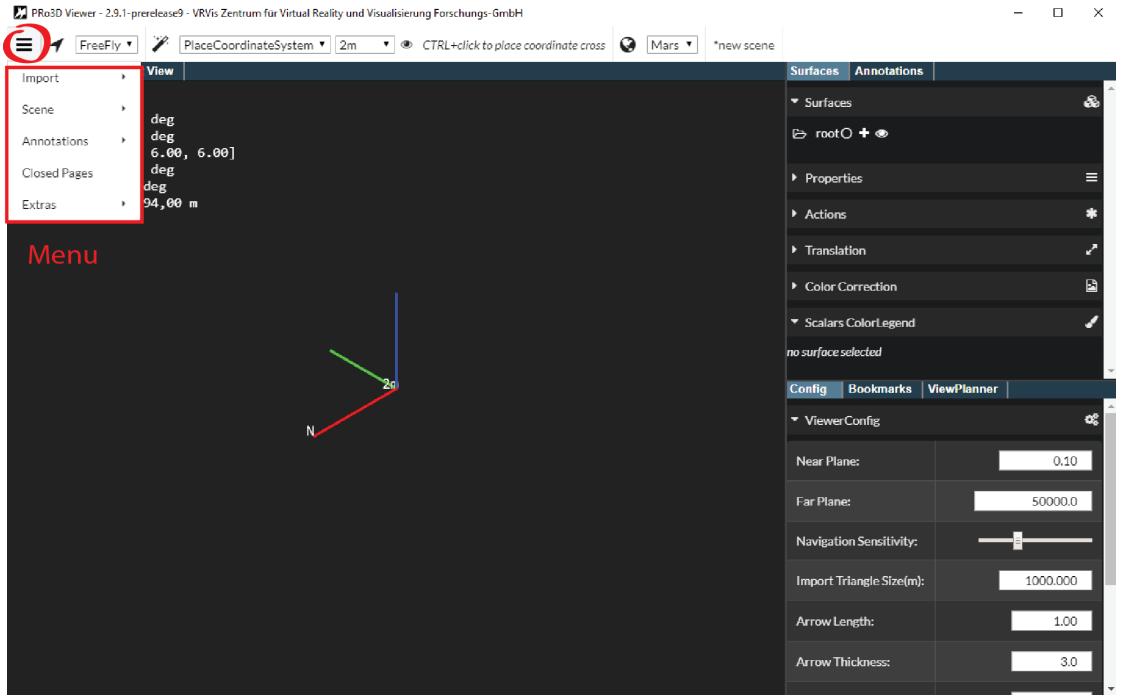


Figure 1: PRo3D Main Menu.

You have two options to start:

- Add one or more surfaces and create a new scene, as described in the next Section 2.1.
- Load an existing scene described in the Section 2.2.

2.1 Add Surface

To add a new surface move the mouse to the “Import” tab in the menu (Figure 1). Click on the “OPC” tab to choose a surface (with “OBJ” you can import object files). This opens the “Select Folder” window where you can choose one or more folder containing opc files as shown in Figure 2. Click the “Select Folder” button to confirm your selection. The surface is loaded into the viewer and listed in the “Surface” page in the right part of the window as shown in Figure 5, part A. You can add more surfaces in the same way.
Each surface has a little context menu below the surface’s name in the list (Figure 5, B). Click the “FlyTo” button to see the surface in the main window. To see the surface’s properties (Figure 5, C) click on the appropriate name in the list.
Finally, click “Scene -> Save as...” in the menu, name the scene and press the “Save” button (Figure 3) to save the surfaces and your settings in a “.scn”

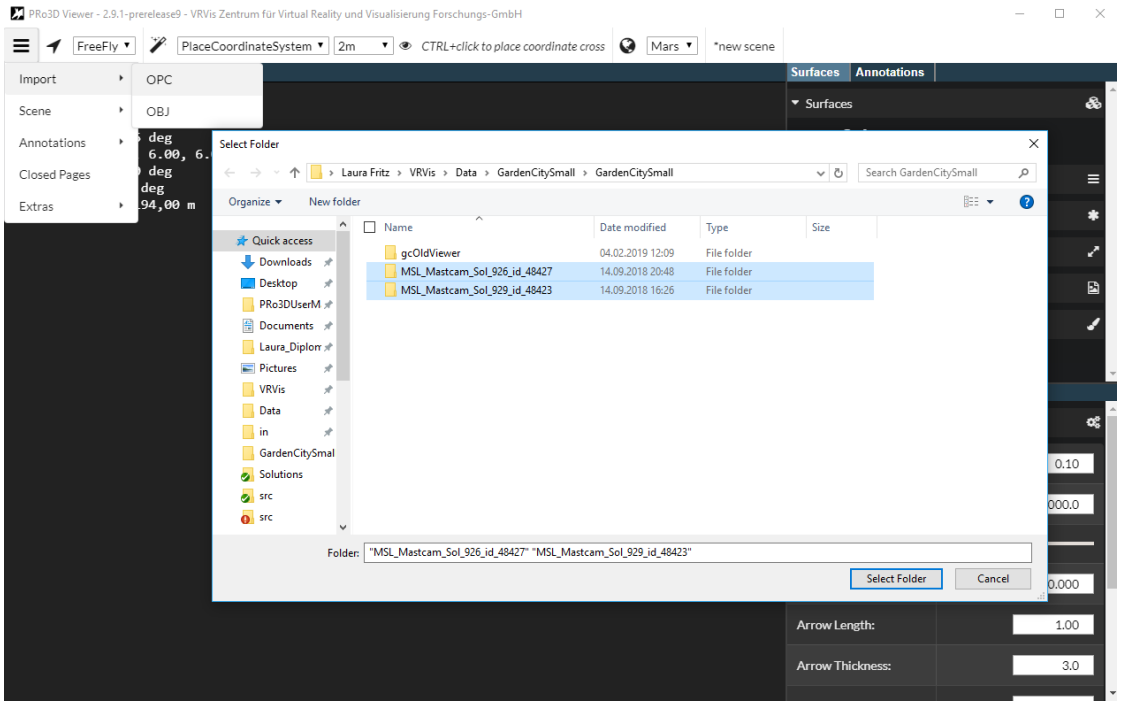


Figure 2: Add new surfaces to the scene.

file. The PRo3D viewer will load the scene automatically next time you start the viewer.

2.2 Load Scene

Load an existing scene by selecting “Scene -> Open” in the **Scene** tab in the start menu (Figure 1). Select the scene xml file in the directory of your choice and confirm your selection (Figure 4). Then the scene is loaded (Figure 5).
You can also load recent scene files with the “Scenes -> Recent” tab in the Scene tab of the menu . By hovering with the mouse over the tab, a list of recent scenes opens. Click on the required scene name to load the scene.

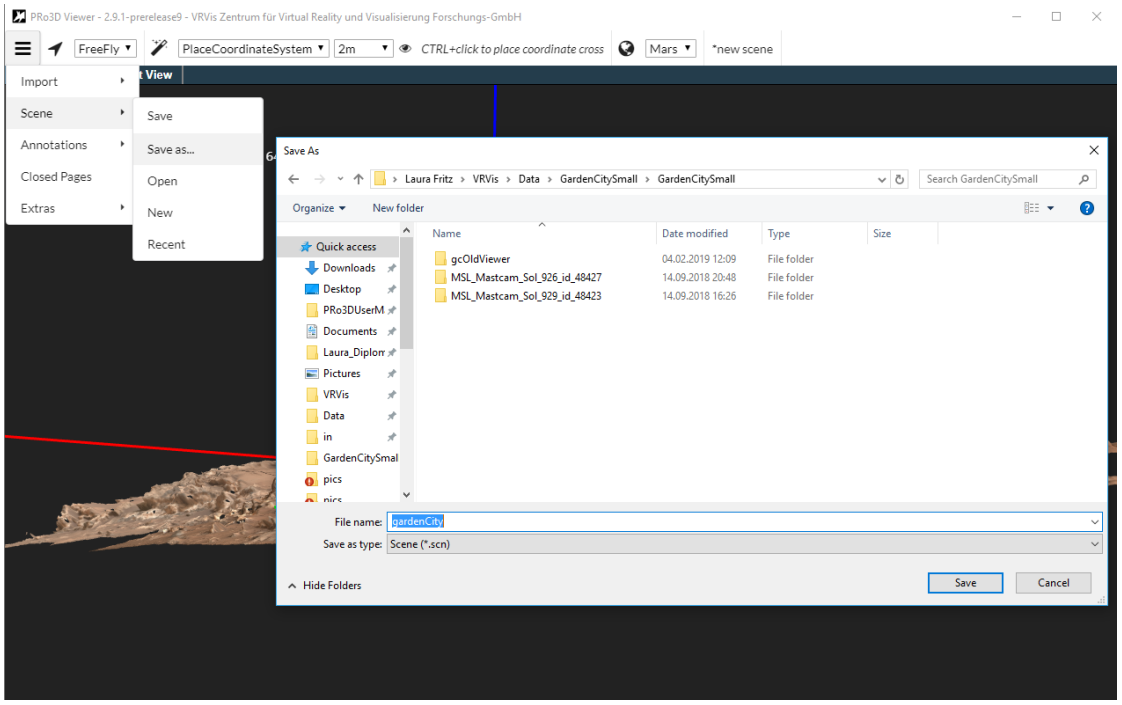


Figure 3: Save the surfaces and settings as scene.

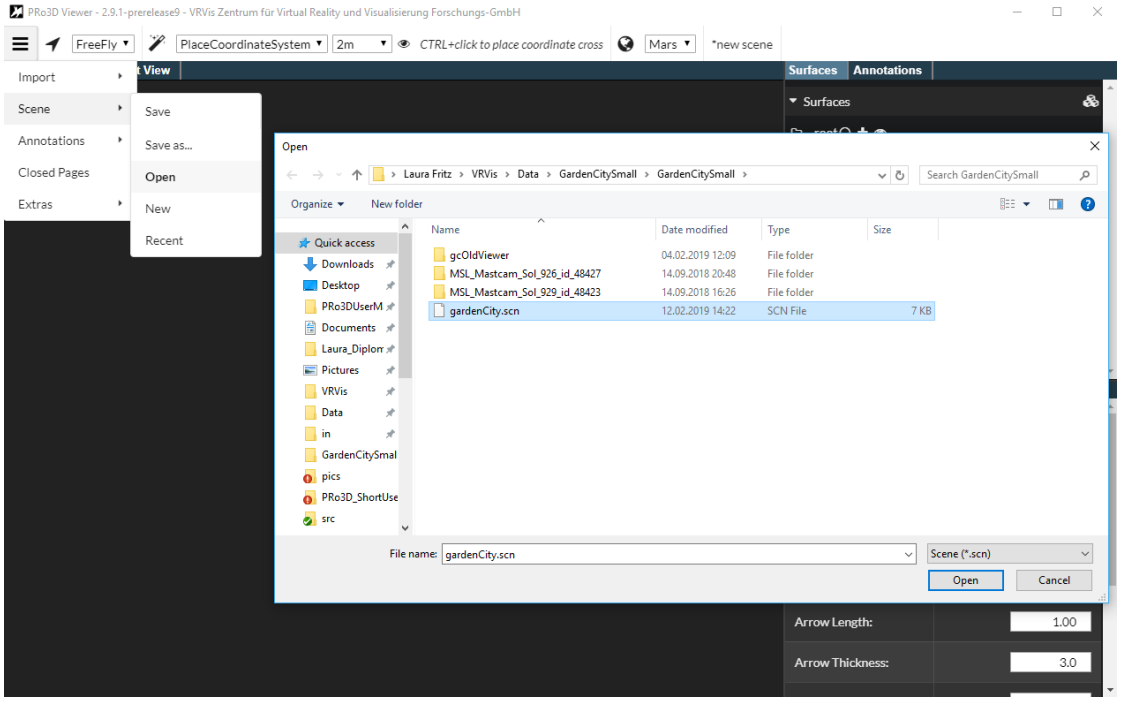


Figure 4: Open a scene with either the “Scene -> Open” or the “Scene -> Recent” tab in the scene tab of the menu.

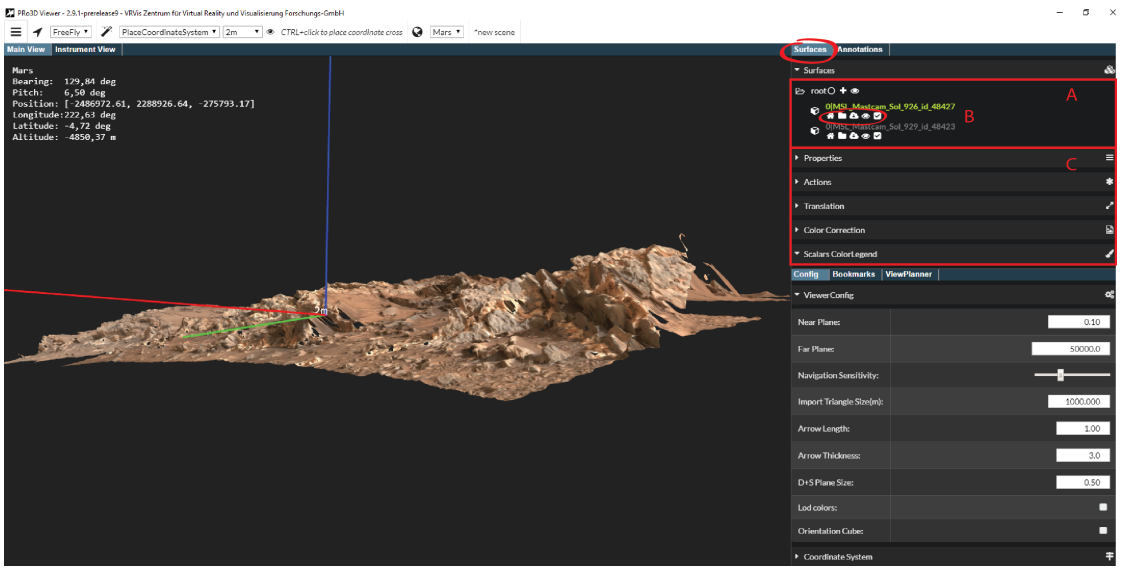


Figure 5: Loaded surface with open surface features (right).

3 Viewer Actions

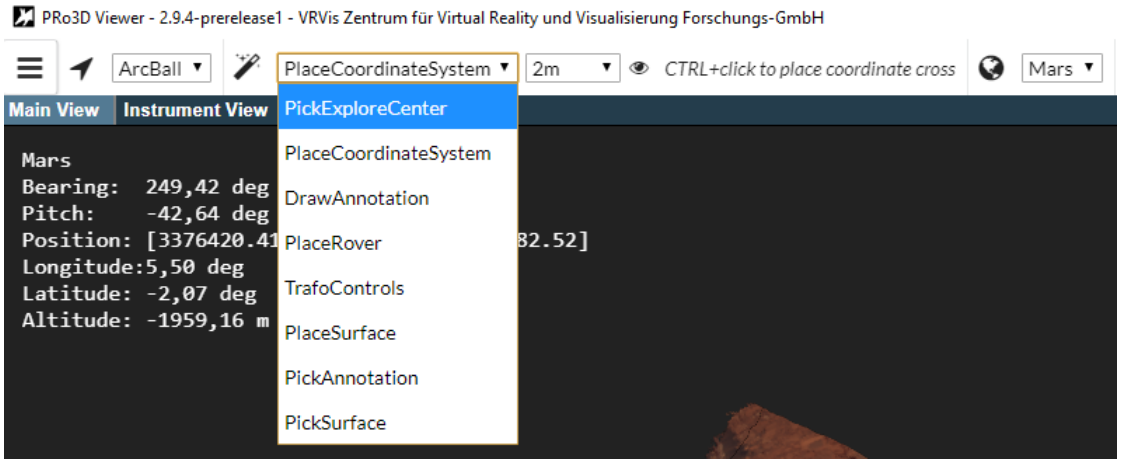


Figure 6: Viewer Actions.

Once the surface is loaded there are different actions to choose (Figure 6).

3.1 Pick Explore Center

The “PickExploreCenter” action concerns the **ArcBall** navigation mode. There are two navigation modes (Figure 7):

- **FreeFly** and
- **ArcBall**.

3.1.1 FreeFly

The Free Fly Mode is the standard 3D fly-through navigation, as for instance, in terrain visualization. WASD controls forward/backward and strafing movement, while the user can change the camera’s orientation by holding the LMB (= **L**eft **M**ouse **B**utton). Zooming in and out (forward/backward movement) is performed by turning the mouse-wheel or holding down the RMB (= **R**ight **M**ouse **B**utton). Additionally, the camera can be panned by holding down the middle mouse button.

3.1.2 ArcBall

When the viewer is in ArcBall mode the camera can be rotated around the explore center by holding down the LMB. Panning and strafing as possible as described above, but be aware that this moves the explore center (otherwise panning would break the view matrix of the camera). To set a new explore point, make sure the “PickExploreCenter” action is active (as shown in Figure 7) and press CTRL + LMB on the surface. The explore center is indicated by a pink dot.

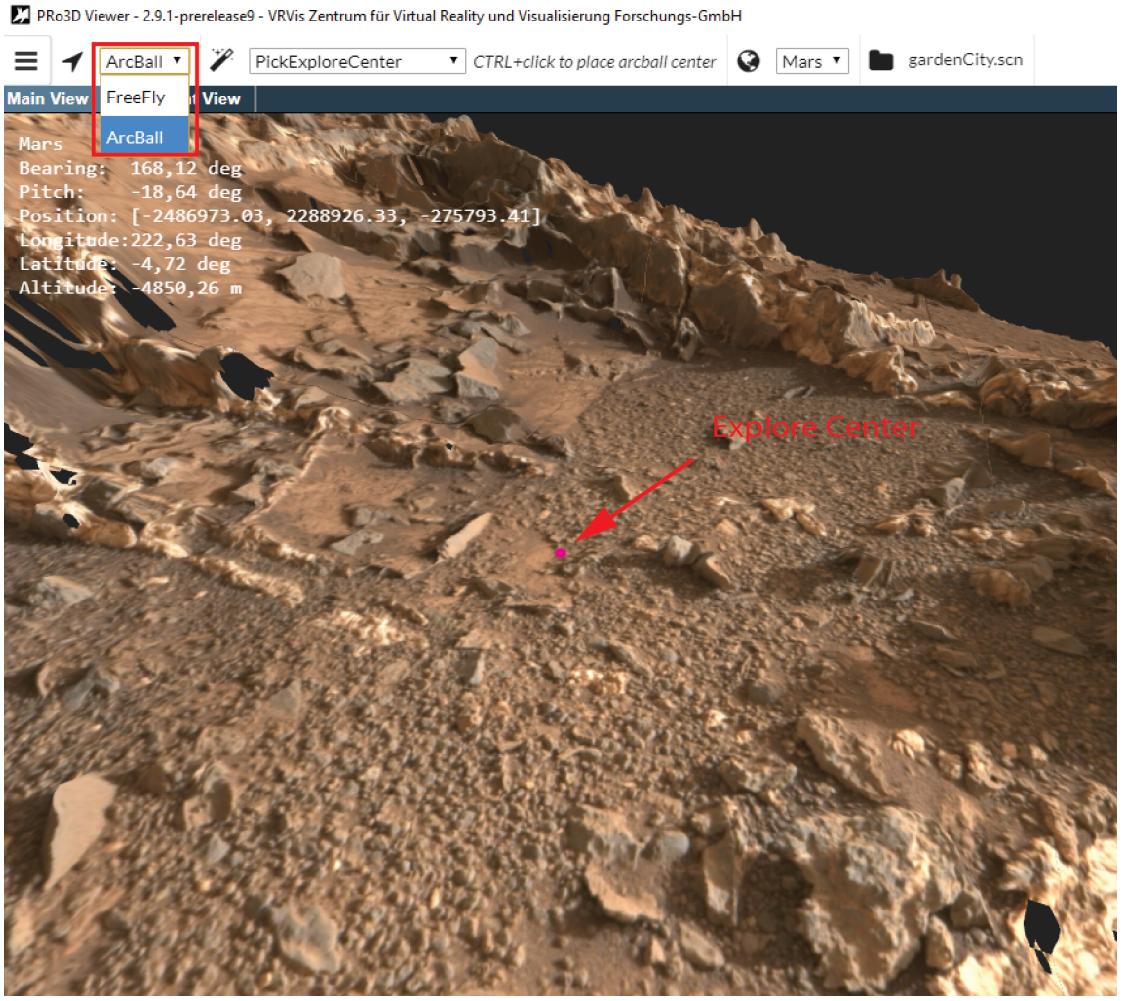


Figure 7: ArcBall navigation. Explore center is set with CTRL + LMB and indicated by a pink dot.

Forward and backward movement is performed either by the keys W/S, rotating the mouse wheel, or clicking RMB.

3.2 Place Coordinate System

To set the coordinate system pick “PlaceCoordinateSystem” in the actions drop down menu, select a unit of measurement and press CTRL+LMB to pick a point on the surface. This marks the position as shown in Figure 8. Initially the up vector’s (blue) direction is set in the positive z-direction and the north vector’s (red) in the positive y-direction. You can manipulate the north vector manually for different data. The surface can be translated along the three directions of the coordinate system, as described in Section 4.1. The up- and the north vector are used for the projection measurements. The north vector is further relevant for bearing measurements and the rover placement in the View Planner. The value for manipulating the north vector is shown in the Viewer Configuration described in Section 4.4.

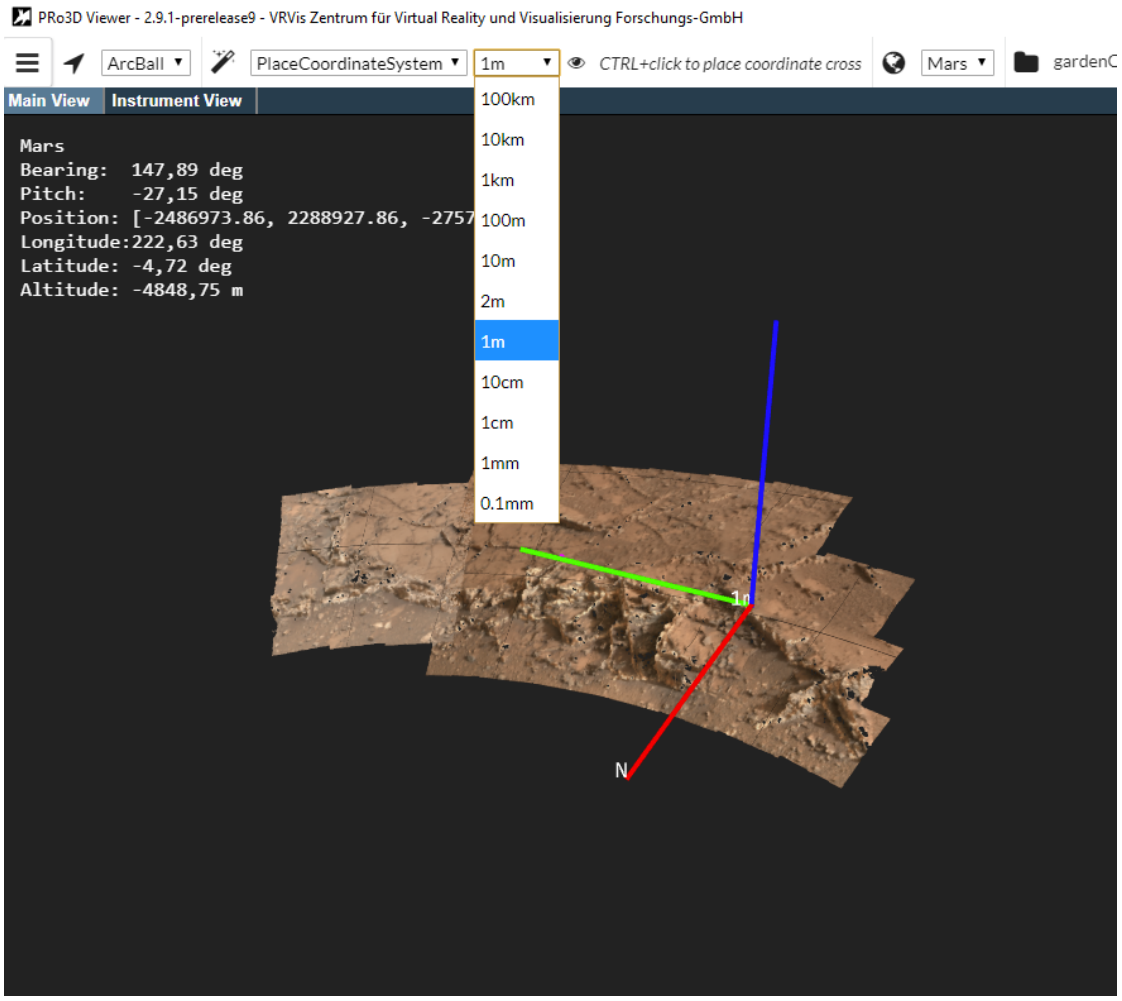


Figure 8: Coordinate System with scale bar functionality.

3.3 Draw Annotation

Figure 9 shows the settings for drawing annotations. First set “DrawAnnotation” in the actions menu (A in Figure 9). Then you can choose one of the following annotation modes (B in Figure 9):

- *Point*: A single point measurement on the surface.
- *Line*: You can pick two points on the surface. The connecting line depends on the projection mode explained in the table below.
- *Polyline*: An arbitrary number of points on the surface can be picked. The connecting line segments depend on the projection mode. The polyline is finished by pressing Enter.
- *Polygon*: An arbitrary number of points on the surface can be picked. The connecting line segments depend on the projection mode. The region of interest is closed and finished by pressing Enter.

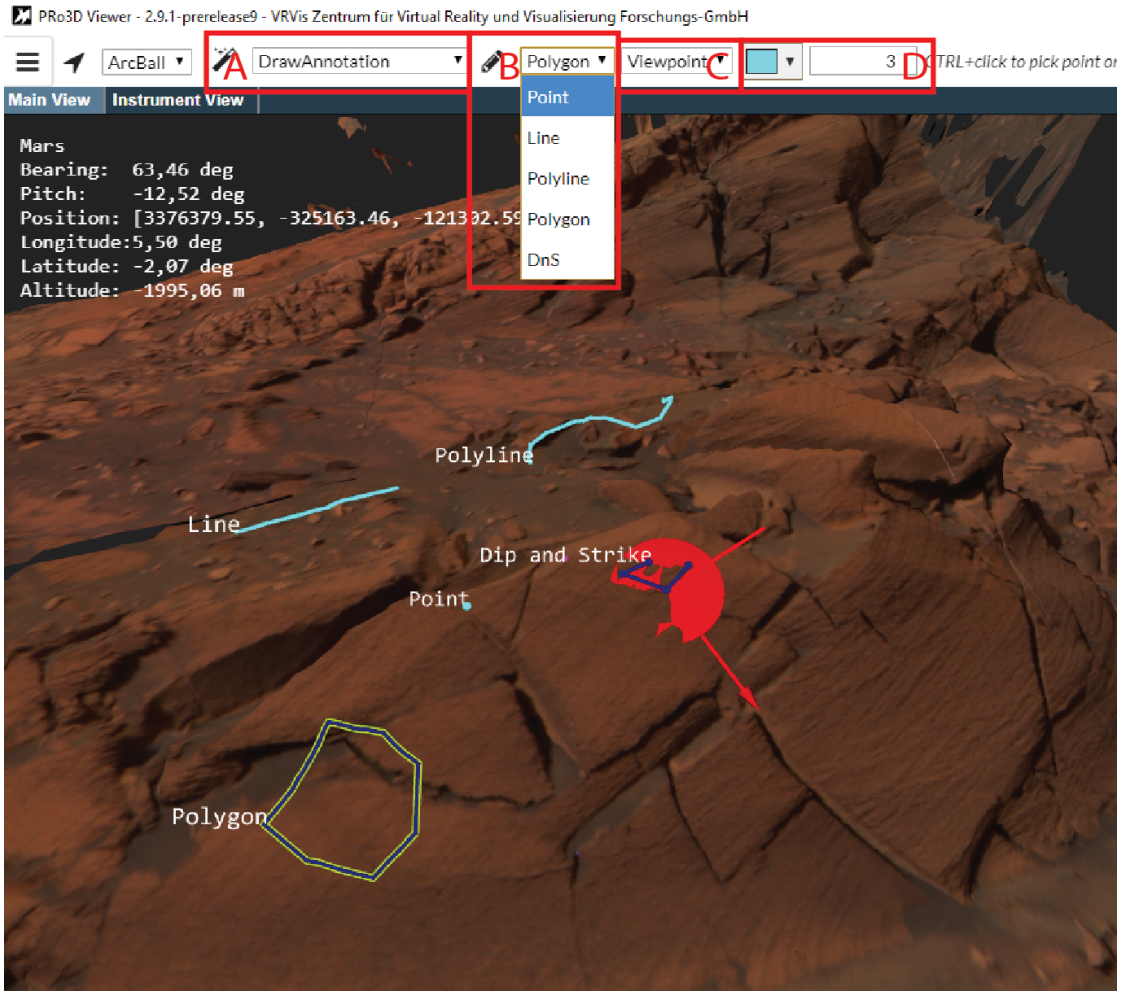


Figure 9: Annotation settings: **A**: set viewer action to "DrawAnnotation. **B**: select annotation mode. **C**: select projection. **D**: choose color and thickness.

- *DnS*: (= *Dip and Strike*) A polyline onto a surface, e.g. alongside a rock layer can be picked. After clicking Enter, a plane is fitted (least squares) to this polyline (blue), which is then intersected with a horizontal plane, which gives us the so-called strike vector (red). This vector represents the direction within the plane with the least inclination and orthogonal to it is the dip vector (green) which shows the direction of highest inclination as shown in Figure 9.

To pick an annotation point on the surface press CTRL+LMB.
For all annotations (except Point) you can choose between "Linear", "ViewPoint" and "Sky" projection which determines the direction of the picking ray (C in Figure 9):

- *Linear*: produces straight line segments as point-to-point connections with linear interpolation between them, no actual projection is performed (Figure 10, blue line). This is practical for line-of-flight distance measurements or measuring the height of a cliff or determining its slope.

- *ViewPoint*: between two points we sample the space by shooting additional rays to intersect with the surface, in this case along the view direction (Figure 10, green line). This is helpful to measure details in nooks and crannies of a rough surface and is the typical way to go for geological measurements.
- *Sky*: the same sampling happens as for the viewpoint projection but this time the rays are shot along the scenes up-vector (Figure 10, pink line). This mode is useful for geographical measurements to estimate the length of a path through a crater.

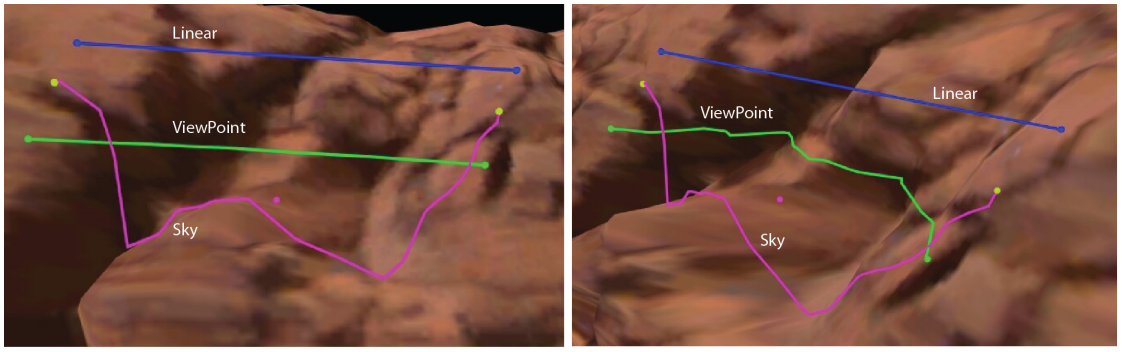


Figure 10: Three Lines taken from the same camera position (left) with the same camera settings but with different line projections.

Finally you can set the color and thickness of the annotation (D in Figure 9). Section 4.2 shows how to maintain, group and edit existing annotations.

3.4 Pick Annotation

You have two possibilities to select an annotation.

1. You can select an annotation by clicking on the annotation's name in the annotation's listing described in Section **??**.
2. Or you set "PickAnnotation" in the actions menu, press CTRL+LMB and pick the annotation in the main view.

Then the annotation turns its color to green in the listing and gets a green border in the main view as shown in Figure 9, where the Polygon is selected. Once selected, you can explore and adjust the annotation's properties (see Section 4.2).

3.5 Pick Surface

You have two possibilities to select a surface.

1. You can select a surface by clicking on the surface's name in the surface's listing described in Section 4.1.

2. Or you set “PickSurface” in the actions menu, press CTRL+LMB and pick the surface in the main view.

Then the surface’s name turns its color to green in the listing. Once selected, you can explore and adjust the surface’s properties (see Section 4.1).

4 Viewer Features

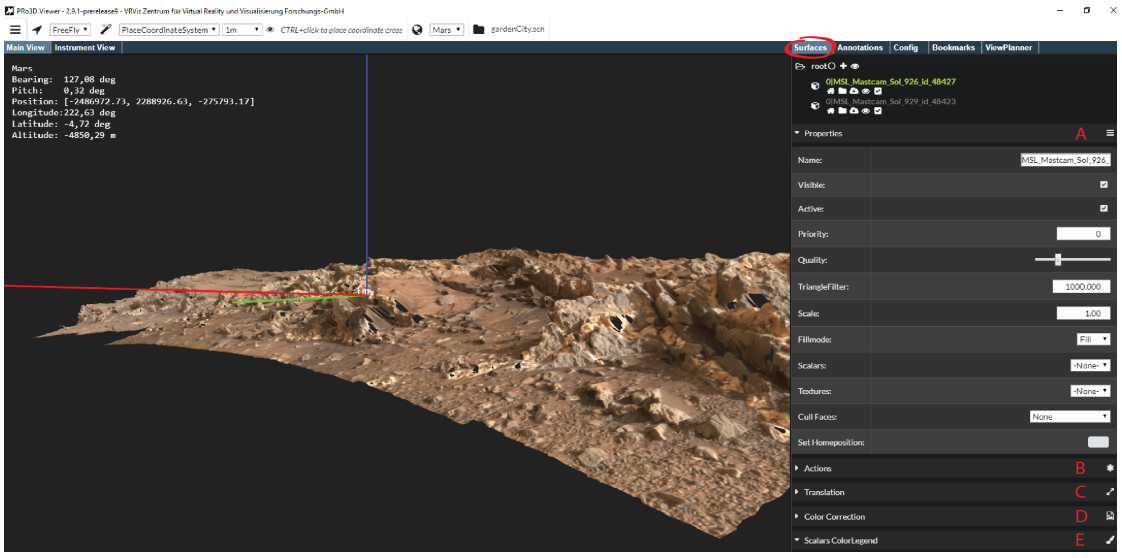


Figure 11: There are pages (right) for each feature in the viewer.

The feature pages show a list of the respective features, the properties of the selected feature from this list and some actions for this feature. For surfaces, annotations and bookmarks it is possible to group them, as described in Section 4.5.

4.1 Surfaces

The listing shows all surfaces in the scene. You can classify them in any group and subgroup layers, described in Section 4.5. You can select a surface by clicking on the surface's name, which turns its color to green. Or you can set "PickSurface" in the actions menu (Figure 6), press CTRL+LMB and pick the surface in the main view. Then you can see the surface's properties in the properties panel and use the actions in the actions panel. It is also possible to select multiple surfaces by clicking the square icon in front of each surface. The selected surfaces have a green square in the list. The multiselection is used to move one ore more surfaces from one group to the active group. Under the surface's name is a little menu:

- *FlyTo*: A click on the button triggers a FlyTo animation.
- *openFolder*: Opens the folder where the scene file resides.
- *Cloud*: Creates new kd-tree files.
- *Toggle Visible*: Toggles the surface visible/invisible.
- *Toggle IsActice*: You can only pick on active surfaces (explore center, annotations, ...).

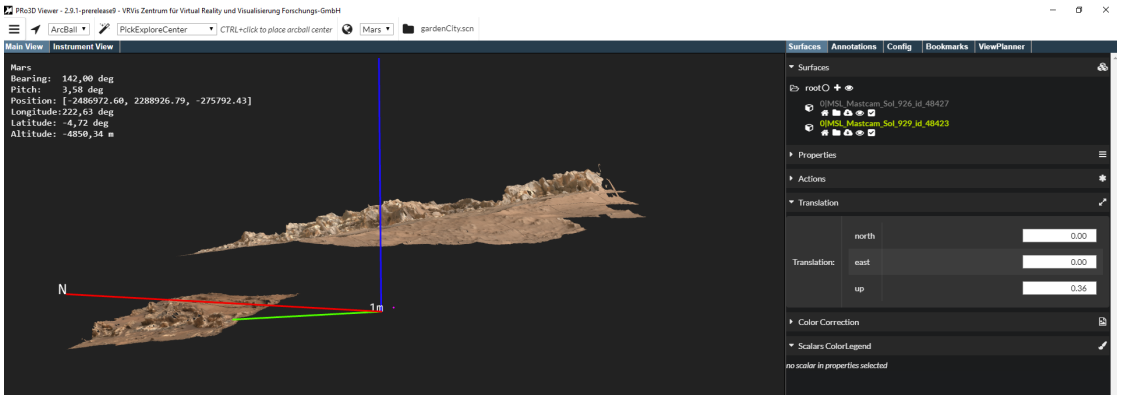


Figure 12: Translation of the selected surface along the axes of the coordinate system.

For each selected surface there are several panels as shown in Figure 11 A to E.
In the surface properties panel (Figure 11 A) are some adjustments possible:

- *Name*: The surface's name. You can change it in the text field and press "enter".
- *Visible*: The surface is visible (checked) or not (unchecked).
- *Active*: The surface is active (checked) or not (unchecked). You can only pick on active surfaces.
- *Priority*: Often multiple surfaces are available for a certain area on the planet surface. These surfaces represent the same piece of ground and typically overlap. This parameter allows to assign a priority to a surface to tell the graphics card which surface should be rendered in front. Lower numbers mean a higher priority in rendering, with 0 being the highest priority. You can also give the highest priority surface a ranking of, for instance, 0.1 (= 10cm) to make annotations more visible. The priority can be dynamically changed via the surface properties so you can try out what works best.
- *Quality*:
- *TriangleFilter*: Excludes triangles with edges bigger than the entered value.
- *Scale*:
- *FillMode*: You can switch between solid/ wireframe/ point rendering of the geometry.
- *Scalars*: Select an attribute layer.
- *Textures*: Select a texture layer.
- *Cull Faces*:

- *Set Homeposition*: You set a new camera position for FlyTo.

The surfaces actions (Figure 11 B) are described in Section 4.5.2.

You can translate the surface along the north-, east and up axis of the coordinate system, described in Section 3.2. And you can rotate the surface around the up vector of the coordinate system. The Translation panel is shown in Figure 11 B and Figure 12.

NOTE: Translation and Rotation are only working for surfaces. Annotations, Rovers, the Coordinate System, ... will NOT move with the surface. You have to transform the surface first!

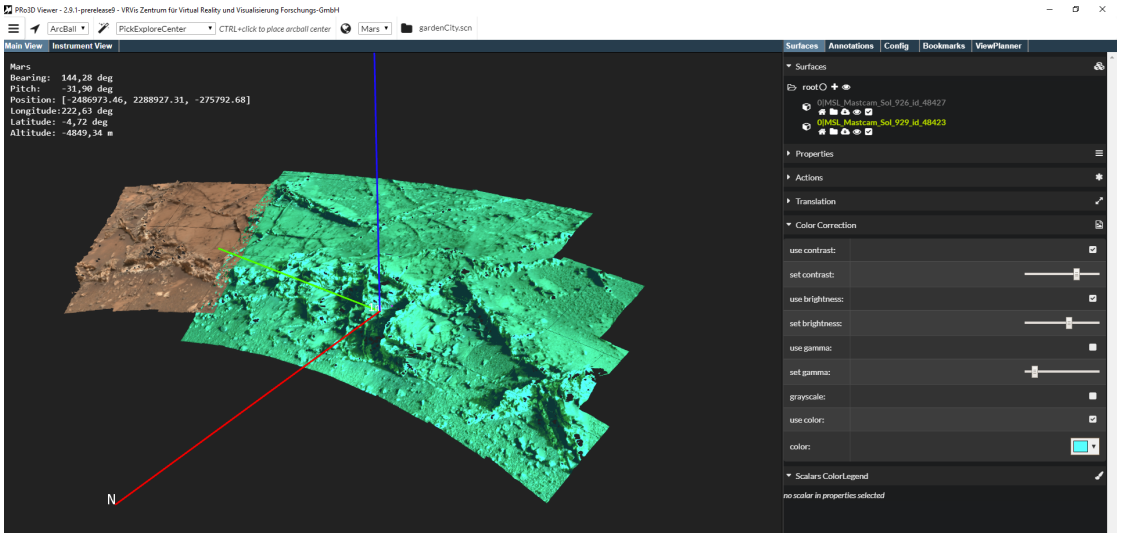


Figure 13: Contrast-, brightness- and color filters applied on the selected surface.

You can apply different color correction filter on the selected surface (Figure 13).

Within the effort of including more and more meta data for a surface we included the so-called SurfaceAttributes(see Figure 14), which are specified in an .opcx file carrying the name of the respective surface. For now, these surface attributes mainly contain additional layers, which can either be a texture layer or an attribute layer. Texture layers are just alternative texture maps that can be mapped onto the surface, such as images from different filters or even sensors (spectral image). Attribute maps on the other hand present an additional value for each position of a surface. If a surface has an .opcx file attached its layers are listed and can be selected in the *Scalars* and the *Textures* combo boxes as part of the surface's property control (Figure 11). In the Scalars ColorLegend panel, the color legend can be adjusted (Figure 14).

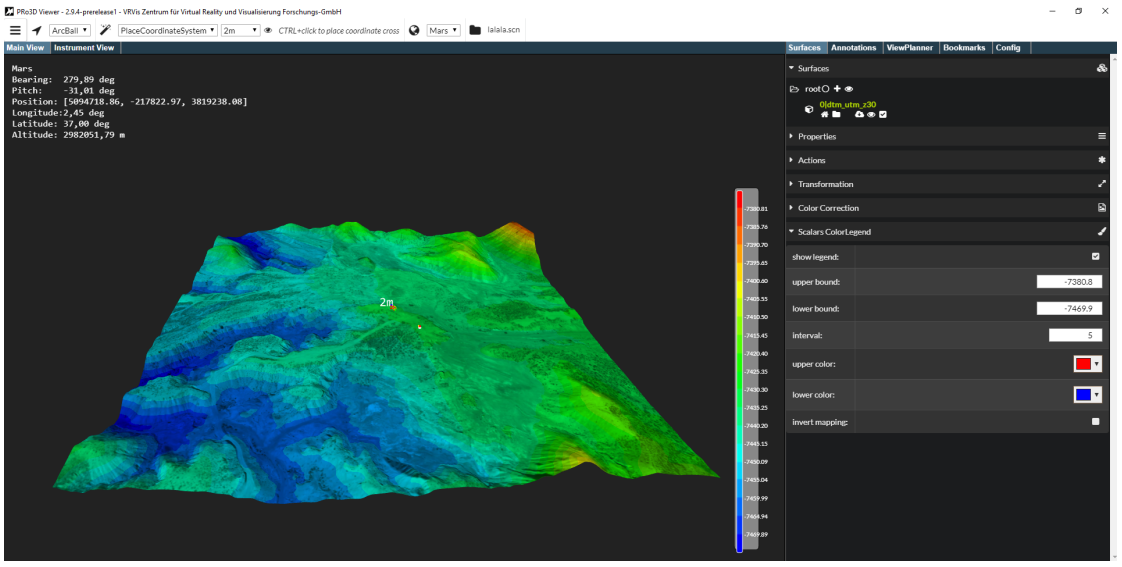


Figure 14: A height map visualized by false color mapping.

4.2 Annotations

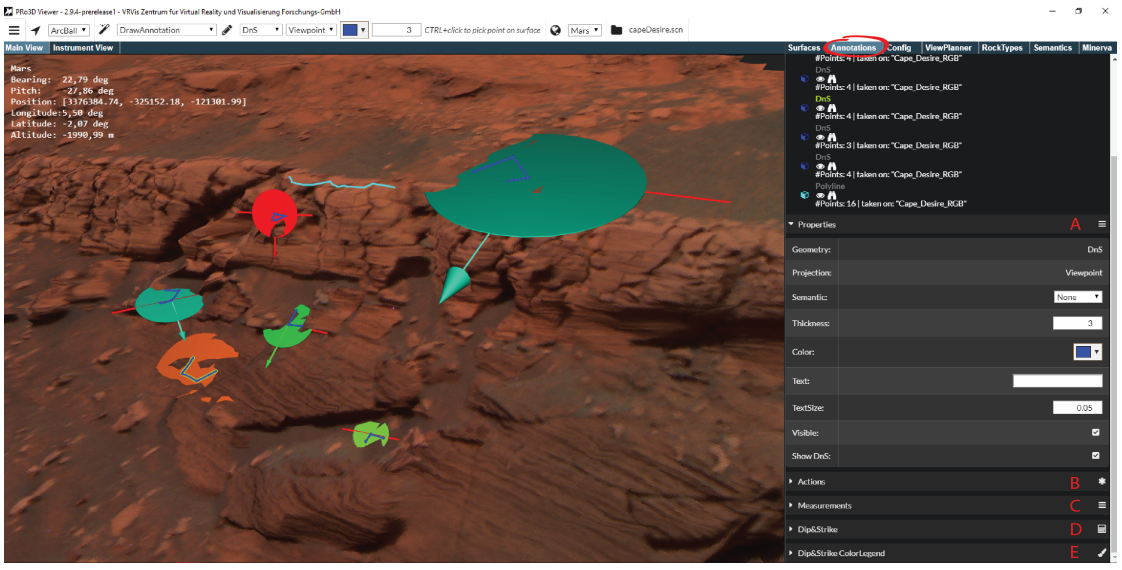


Figure 15: The annotations page.

The listing shows all annotations in the scene. You can classify them in any group and subgroup layers, described in Section 4.5. You can select an annotation by clicking on the annotation's name, which turns its color to green. Or you can set "PickAnnotation" in the actions menu (Figure 6), press CTRL+LMB and pick the annotation in the main view. Then you can see the annotation's properties in the properties panel and use the actions in the actions panel. It is also possible to select multiple annotations by clicking the square icon in front of each annotation. The selected annotations have a green square in the list. The multiselection is used to move one ore more annotations from one group to the active group.

Under the annotation's name is a little menu:

- *Toggle*: Toggles the annotation visible/invisible.
- *FlyTo*: A click on the button triggers a FlyTo animation.

For each selected annotation there are several panels as shown in Figure 15 A to E.
The properties of the selected annotation (click on annotation's name) are shown in Figure 15 A. There you can get information and change some of the settings:

- *Geometry*: Shows the annotation mode (described in Section 3.3). This is not changeable retrospectively.
- *Projection*: Shows the projection which determines the direction of the picking ray (described in Section 3.3, shown in Figure 10). This is not changeable retrospectively.
- *Semantic*:
- *Thickness*: You can change the annotation's line thickness.
- *Color*: You can change the annotation's color.
- *Text*: You can append a note. Write in the text field and press "Enter". The note will appear next to the annotation in the viewer.
- *TextSize*: You can change the textsize.
- *Visible*: The annotation is visible (checked) or not (unchecked).
- *ShowDnS*: For each annotation with more than three picking points Dip and Strike information (Section 3.3) is available. The DnS is visible (checked) or not (unchecked).

The annotations actions (Figure 15 B) are described in Section 4.5.2.

The measurements tap (Figure 15 C) contains some information:

- *Position*: Shows the position (only for point annotations).
- *PrintPosition*: Prints the position in the console window.
- *Height*: The height between the annotation's start and end point.
- *HeightDelta*: The height difference between the highest and lowest point of the projected line.
- *AvgAltitude*: The average altitude.
- *Length*: The sum of direct distances between the picking points.
- *WayLength*: The sum of projected distances between the picking points.

- *Bearing*: The annotation's bearing.
- *Slope*: The annotation's slope.
- *Vertical Distance*: The vertical distance between the annotation's start and end point in relation to the up vector of the coordinate system.
- *Horizontal Distance*: The horizontal distance between the annotation's start and end point in relation to the north- and right vector of the coordinate system.

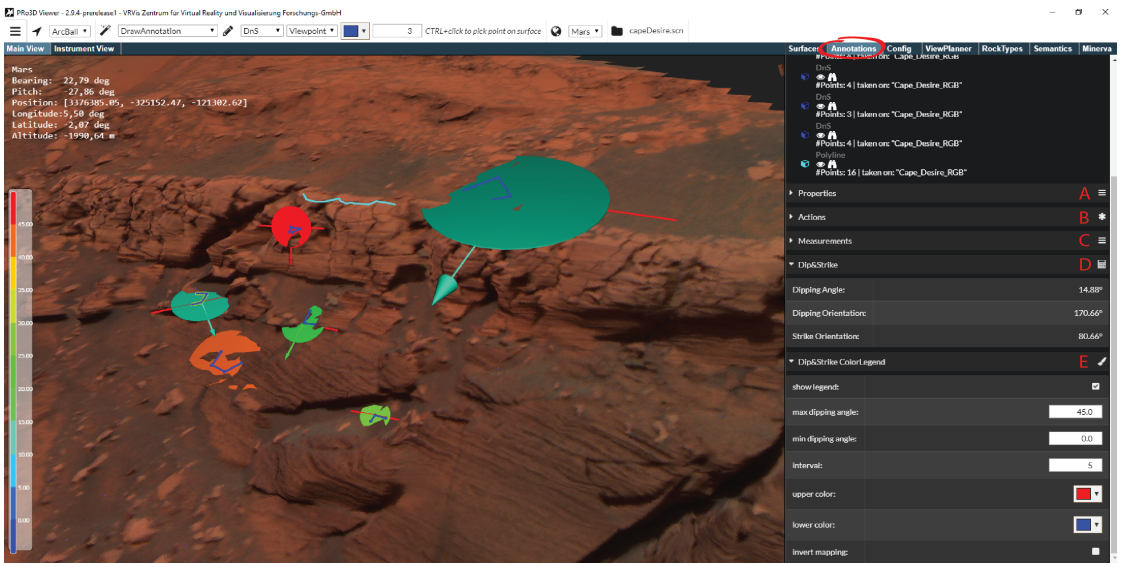


Figure 16: Colorcoding of the Dip&Strike annotations according to their dipping angles.

Measurements for the dip and strike annotations are shown in the Dip&Strike tab (Figure 15 D and Figure 16 D).

The color coding of the DnS annotation's discs and arrows is specified by the dipping angle. The color legend's parameters can be adjusted in the Dip&Strike ColorLegend tab shown in Figure 16 E.

4.3 Bookmarks

Bookmarks enable the user to record a certain camera viewpoint. To add a new bookmark click the "+" button on top of the page. The new bookmark is added to the active group in the bookmarks listing. To view the bookmark's properties and actions click on the bookmark's name. Clicking the "house" button beside the bookmark's name triggers a FlyTo. For multiselection click on the bookmark's square icons.

4.4 Viewer Configuration

To edit the viewer properties, select the "Config" page.

4.4.1 ViewerConfig

A set of major viewer properties can be adjusted:

- *Near/Far Plane*: The near- and the far clipping plane are automatically adjusted according to the data to be rendered. The set values are shown in the config panel and can be adjusted afterwards.
- *Navigation Sensitivity*: The navigation sensitivity can also be adjusted by PageUp and PageDown keys.
- *Arrow Length/Thickness*: The arrow length and thickness is set for up- and north vectors, dip and strike vectors and the up- and lookAt vectors in the rover view planner.
- *D+S Plane Size*: The dip and strike measurements plane size, described in Section 3.3 (Figure 9).
- *Min/Max Dipping Angle*: The dip and strike measurements dipping angle range. The dipping angle is coded into the color of the disc and arrow of a measurement (Figure 9).
- *Lod Colors*: The different levels of detail of the surface geometry can be colored in different shades of red. This helps to evaluate the export of OPC data.

4.4.2 Coordinate System

The coordinate system menu shows the position, Up- and North Vector of the coordinate system described in Section 3.2, shown in Figure 8. The Up- and the North Vector are used for the projection measurements (Figure 9). Initially the Up Vector's direction is set in the positive z-direction and the North Vector's in the positive y-direction. But you can manipulate the Up Vector manually for different data. Both vectors are computed automatically with picking of a new position for the coordinate system. The north vector is further relevant for bearing measurements.

4.4.3 Camera

The Camera submenu shows the Location, Forward- and Sky vector of the main camera.

4.5 Grouping

Grouping is possible for surfaces, annotations and bookmarks. The "root" group is the highest level where you can add leafs and subgroups. Each group has a context menu:

- *Set Active*: The active group gets the new leaf. Per default the “root” is active.
- *Add Group*: Adds a new and empty subgroup.
- *Toggle Group*: Sets all leafs in this group and its subgroups invisible.

4.5.1 Group Actions

- *Remove*: Removes the group with all its leafs and subgroups.
- *Clear*: Removes all leafs and subgroups from group but retains the empty group.
- *Selection: Move*: Moves all selected leafs (green squares) to the active group.
- *Selection: Clear*: Clears the selection (the leafs were not removed).

4.5.2 Leaf Actions

- *Remove*: Removes the leaf.
- *Selection: Move*: Moves all selected leafs (green squares) to the active group.
- *Selection: Clear*: Clears the selection (the leafs were not removed).

5 Minerva



Figure 17: The Minerva Application. The Main View (left) shows the surface data with the products. There are different interaction possibilities for the products in the Minerva tab (right). A: Visplore communication, B: Query App to reduce the shown products by the mean of different criteria, C: Product listing, D: Properties of the last selected product, E: Surface tab.

5.1 Start Minerva

Before you start the Minerva Application make sure that the **dump.csv** file is in the **`Release\netcoreapp2.0\MinervaData`** folder. Then start the viewer by clicking the PRo3D.exe as described in Section 2. The Minerva data from the dump.csv file is loaded and the products are listed in the “Minerva” tab as shown in Figure 17 C. Load the surface data as described in Section 2.1. How to explore and adjust the surfaces is described in Section 4.1. You can save the scene as described in Section 2.1.

5.2 Minerva Features

5.2.1 Visplore

Visplore is a powerful visual analytics tool that allows the handling of huge and heterogeneous data and provides different perspectives onto the data that help finding causalities and allow an easy examination of the data’s plausibility. For this project an interface was implemented that allows an interconnected workflow between PRo3D and Visplore. To communicate with Visplore use the tree buttons shown in Figure 17 A:

- **Connect:** Connect to Visplore.

- **Send Selection:** Send the selected products to Visplore.
- **Snapshot:** Send a snapshot of the MainView to the Visplore MapView.

For more information how to use Visplore for Minerva, please refer to the Visplore manual.

5.2.2 QueryApp

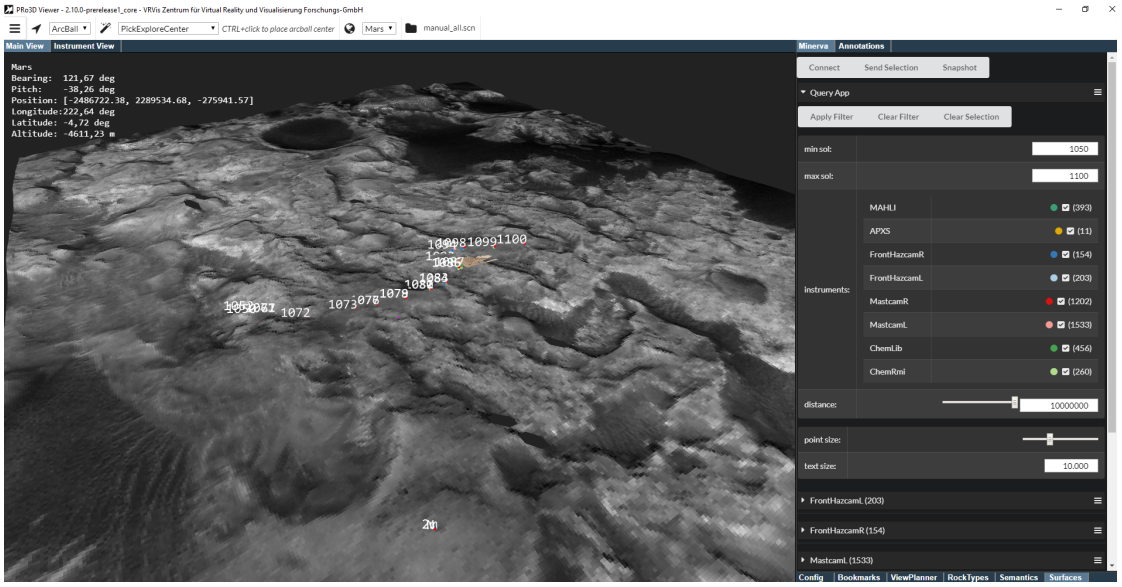


Figure 18: The Query App (right).

The Query App provides the possibility to cut down the number of shown products by the mean of different criteria. These criteria are:

- **min- and max sol:** All products with sol numbers in the given range are shown.
- **instruments:** Only the products with checked instruments are shown.
- **distance:** The distance between the camera location and the shown products is less than the entered distance value.

You can use one ore more criteria and press the **“ApplyFilter”** button in top of the Query App menu. Now only the filtered products are shown. Change the queries and click “ApplyFilter” again to change the shown products. To get all products click the **“ClearFilter”** button. You can also change the **point size** of the rendered products in the Main View, and the **text size** of the shown sol numbers.

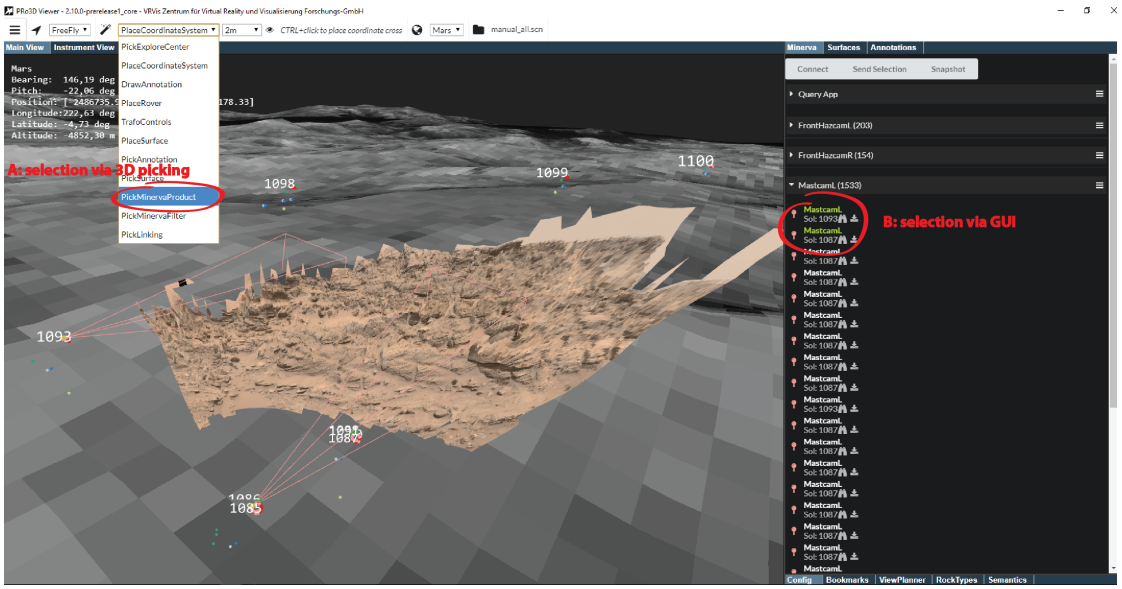


Figure 19: The products are grouped by instruments. A product can either be selected in the listing (B) or in the Main View (A).

5.2.3 Product Listing and Selection

The products are grouped in the gui by their instruments (Figure 17 C) where the first 20 products for each instrument are listed. You can select single products by clicking on the products's name, which turns its color to green. Or you can set "PickMinervaProduct" in the actions menu (Figure 19), press CTRL+LMB and pick the product in the main view. Then you can see the product's properties in the properties panel, described in the next Section 5.2.4.

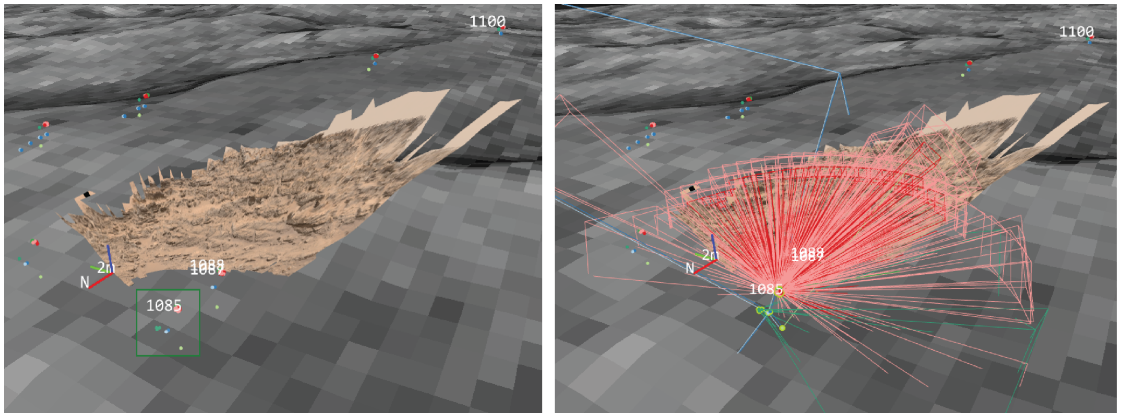


Figure 20: A rectangle is spanned over multiple products (left). All products selected by the rectangle (right).

It is also possible to select multiple products. Press SHIFT+LMB and drag a rectangle (Figure 20 left). When you release the LMB, all products in this area are selected (Figure 20 right).
Under the products's name is a little menu:

- **The Sol Number**

- **FlyTo**: A click on the button triggers a FlyTo animation.
- **GetTif**: Downloads the product's tif image.

5.2.4 Product Properties

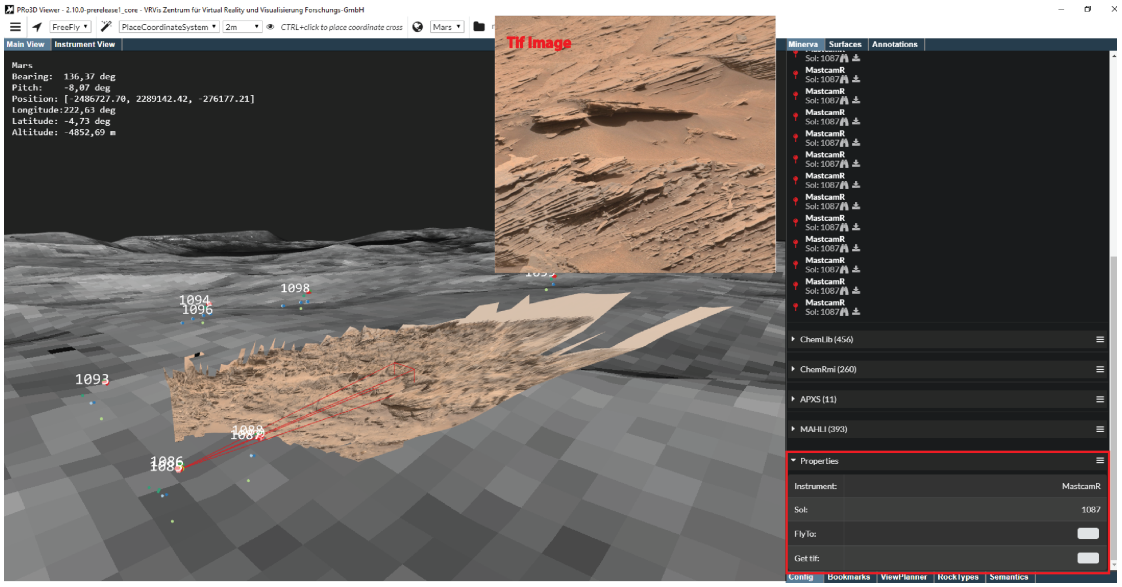


Figure 21: The properties panel shows the sol number and provides interactions for the last selected product.

The properties panel (Figure 17 D and Figure 21) shows

- **The Sol Number**
- **FlyTo**: A click on the button triggers a FlyTo animation.
- **GetTif**: Downloads the product's tif image. The image is stored in the `Release\netcoreapp2.0\MinervaData` folder.

of the last selected product.

5.3 Picking Actions

There are three minerva picking actions in the picking actions drop down menu shown in Figure 23.

- **PickMinervaProduct**: You can set "PickMinervaProduct" in the actions menu to select a product in the Main View as described in Section 5.2.3.
- **PickMinervaFilter**: Select "PickMinervaFilter" in the actions menu and press CTRL+LMB to pick a point on the surface. Then set a distance value with the distance slider in the Query App and click the "ApplyFilter" button. The products where the distance to the selected point is smaller than the selected distance in the Query App are shown (Figure 22).

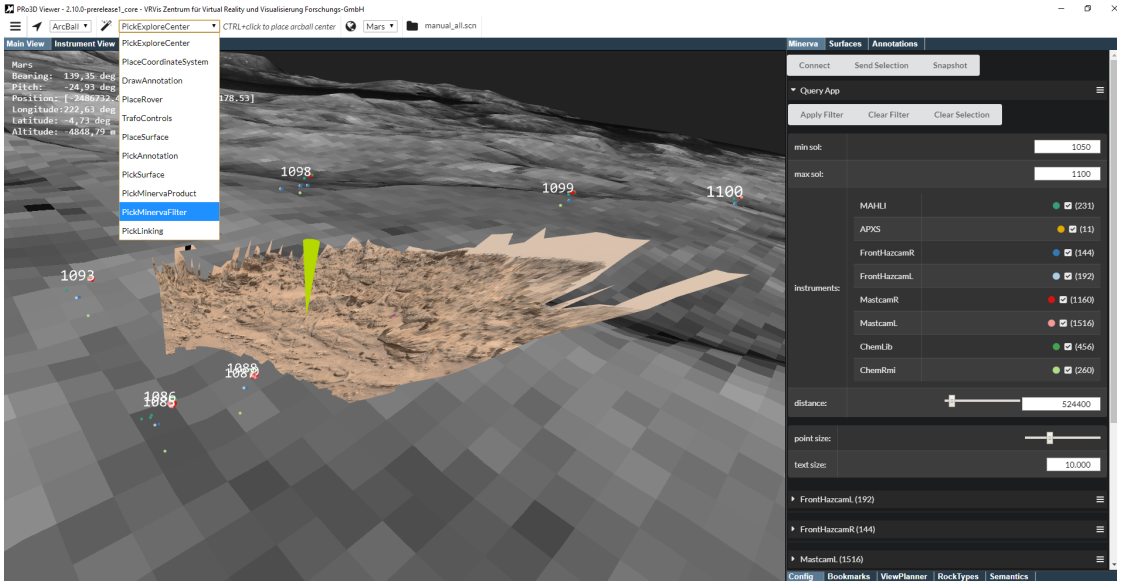


Figure 22: PickMinervaFilter

- **PickLinking**: Select “PickLinking” in the actions menu and press CTRL+LMB to pick a point on the surface. All camera frustums of the products that captures the selected point are shown (Figure 23).

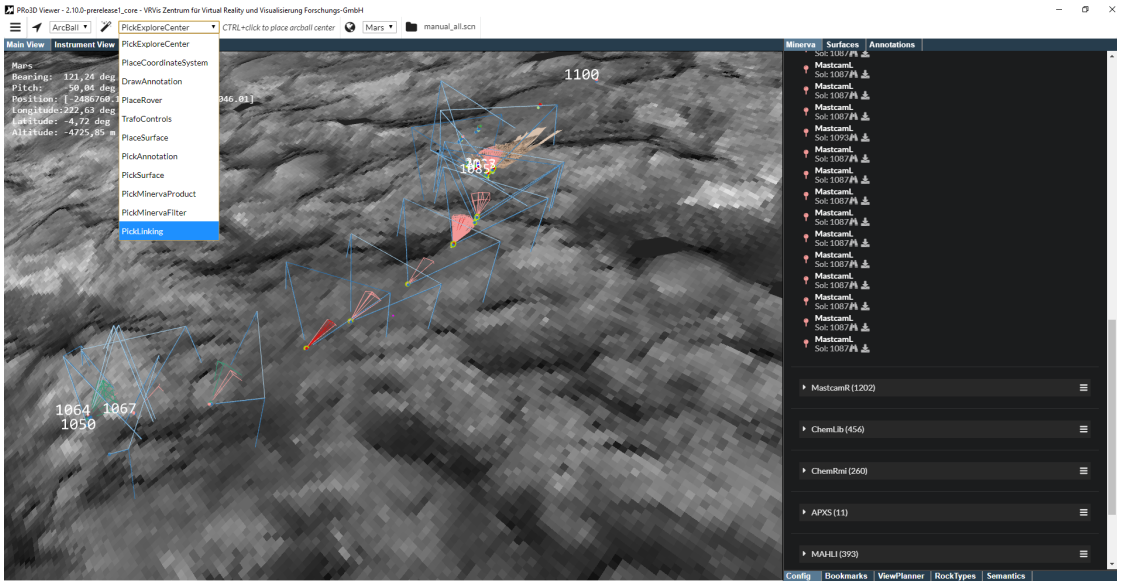


Figure 23: PickLinking